Project Timeline — NeuroShell Core Framework (R26-CS-018)

NeuroShell Core Team Cyber Operations Research Group
MSc Cybersecurity Research Project R26-CS-018

Revision History

Version	Date	Author	Changes
1.0	2026-09-05	NeuroShell Core Team	Initial comprehensive SRS

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I. Project Timeline

NeuroShell Core Framework (R26-CS-018) — MSc Cybersecurity Research Project

This document is the weekly project timeline for the NeuroShell Core Framework, covering all four components as a single system:

- **Component 01 — Intent Recognition Engine (IRE):** NLU gateway converting natural-language operator commands into validated JSON intent contracts.
- **Component 02 — Dynamic Planner RAG Engine:** Retrieval-Augmented Generation planner producing ordered execution graphs and success probabilities.
- **Component 03 — Adaptive Execution and Error Recovery Engine (AEERE):** Self-healing runtime engine categorizing errors and executing recovery strategies.
- **Component 04 — AI Vulnerability Analysis Engine (AVAE):** Scans, parses, enriches (NVD / CISA KEV / FIRST EPSS), predicts exploitability (XGBoost + SHAP) and triages vulnerability findings.

TABLE I
Table 1.

Field	Value
Document ID	NS-PROJ-TIMELINE-001
Version	1.0
Date	2026-09-05
Project	NeuroShell Core Framework — R26-CS-018
Start Date	2026-03-15 (Proposal Submission)
End Date	2026-12-01 (Research Paper Publication Evidence Submission)
Scope	All 4 components (IRE, Planner, AEERE, AVAE)
Purpose	Weekly plan for import into Microsoft Teams Planner

II. 1. Master Milestone Timeline

The following common timeline applies to the 2026 Regular Batch research project (RP):

TABLE II
Table 2.

Action	Deadline
Proposal Submission	15th March 2026
Proposal Presentation	16th to 18th March 2026
Check List submission	13th May 2026
Progress Presentation 1	11th to 13th May 2026
Progress Presentation 2	31st August to 02nd September 2026
Draft Thesis submission	11th October 2026
Website Submission	11th October 2026
Final Check List submission	14th October 2026
Final Presentation and Viva	19th to 21st October 2026
Website evaluation and Logbook submission	19th to 21st October 2026
Research Paper submission	23rd October 2026
Final Thesis Submission	28th October 2026
Research Paper publication evidence submission	1st December 2026

III. 2. Weekly Activity Plan

Weekly granularity is used across the entire project. Weeks are Monday-to-Sunday ranges. Work from **2026-03-15 to 2026-09-05** is already completed and is captured retrospectively; work from **2026-09-06 to 2026-12-01** is scheduled. Component tags used: C1=IRE, C2=Planner, C3=AEERE, C4=AVAE, ALL=whole framework.

TABLE III
TABLE 3.

Week	Date Range	Milestones and Deliverables
W00	2026-03-15 to 2026-03-15	**Proposal Submission** (15 Mar). Submit final project proposal; define scope, objectives, 4-component architecture; create MS Teams channel and Planner board. **Comp: ALL — Completed**

TABLE IV
TABLE 4.

Week	Date Range	Milestones and Deliverables
W01	2026-03-16 to 2026-03-22	**Proposal Presentation** (16-18 Mar). Present proposal to review board; incorporate feedback; refine research questions. **Comp: ALL — Completed**
W02	2026-03-23 to 2026-03-29	Literature review: LLM security agents, prompt injection defenses, RAG, CVSS/EPSS/KEV; collect IEEE / OWASP / MITRE references. **Comp: ALL — Completed**
W03	2026-03-30 to 2026-04-05	Monorepo scaffolding: apps / shared / infra structure, FastAPI + Poetry environments, docker-compose baseline, git workflow, CI skeleton. **Comp: ALL — Completed**
W04	2026-04-06 to 2026-04-12	Kong API Gateway baseline: declarative routing, rate limiting (60 req/min), CORS policy, header injection (X-Gateway-Version, X-Service-Name). **Comp: ALL — Completed**
W05	2026-04-13 to 2026-04-19	IRE Stages 1-2: input normalizer (NFKC, whitespace, slang standardization) and alias resolver with data/alias_map.json (EternalBlue, Log4Shell, etc.). **Comp: C1 — Completed**
W06	2026-04-20 to 2026-04-26	IRE Stage 3: Ollama integration (gemma4:e4b), prompt engineering, strict JSON generation, thinking-tag and markdown stripping. **Comp: C1 — Completed**
W07	2026-04-27 to 2026-05-03	IRE Stages 4-7: JSON extraction, Pydantic IntentSchema validation, regex validation (IP/CIDR/CVE), ScopeGuard (RFC-1918 + prompt injection rejection). **Comp: C1 — Completed**
W08	2026-05-04 to 2026-05-10	Progress Presentation 1 preparation: demo of IRE /parse through Kong, slides, draft of first check list. **Comp: ALL — Completed**

A. Week of Proposal Submission (Kickoff)

B. Phase A — Proposal and Foundations (Clients C1-C4)

C. Phase B — Progress Presentation 1 and Implementation

D. Phase C — Progress Presentation 2 and Documentation

E. Phase D — Thesis Drafting and Website

F. Phase E — Final Deliverables and Viva

G. Phase F — Research Paper Publication

IV. 3. MS Teams Planner Usage

The companion file `MS_Teams_Planner_Import.csv` contains one row per weekly task plus the milestone tasks, ready for import into Microsoft Teams Planner. Columns are: Task Name, Bucket Name, Start Date, Due Date, Progress, Priority, Notes.

- **Buckets** group the plan into Proposal, Phase A Foundations, Phase B Implementation, Phase C Progress 2 and Docs, Phase D Thesis and Website, Phase E Final and Viva, Phase F Publication.
- **Progress** uses Completed (2026-03-15 to 2026-09-05) and Not started (2026-09-06 onwards); switch each task to In progress / Done as work is completed each week.
- **Assignees** can be added in Planner per component owner (C1: IRE, C2: Planner, C3: AEERE, C4: AVAE).

TABLE V
TABLE 5.

Week	Date Range	Milestones and Deliverables
W09	2026-05-11 to 2026-05-17	**Check List submission (13 May); Progress Presentation 1 (11-13 May).** Present IRE pipeline; submit first check list. **Comp: ALL — Completed**
W10	2026-05-18 to 2026-05-24	Planner RAG design: PlanStep / GeneratedPlan schemas, vector store selection, RAG context ingestion. **Comp: C2 — Completed**
W11	2026-05-25 to 2026-05-31	Planner RAG build: context retrieval, query construction, tool selection, command synthesis, safety filter and command validation. **Comp: C2 — Completed**
W12	2026-06-01 to 2026-06-07	AEERE design: error taxonomy (timeout, resource, validation, authentication, network), recovery strategy selection (retry, fallback, circuit breaker). **Comp: C3 — Completed**
W13	2026-06-08 to 2026-06-14	AEERE build: RecoveryPlan model, /process-error endpoint, execution latency and outcome tracking. **Comp: C3 — Completed**
W14	2026-06-15 to 2026-06-21	AVAE scanner parsing: Nmap, Nikto, Gobuster and generic parsers; ParserRegistry dispatch; EntityCollection output. **Comp: C4 — Completed**
W15	2026-06-22 to 2026-06-28	AVAE threat intelligence: NVD client, CISA KEV feed load, FIRST EPSS batch client, SQLite CVE cache. **Comp: C4 — Completed**
W16	2026-06-29 to 2026-07-05	AVAE data pipeline: CCEE dataset ingestion, 12-feature engineering, build/merge/train/test CLI scripts. **Comp: C4 — Completed**
W17	2026-07-06 to 2026-07-12	AVAE ML model: XGBoost training (grid search, 5-feature runtime contract), SHAP explanations, model artifact and training medians. **Comp: C4 — Completed**
W18	2026-07-13 to 2026-07-19	AVAE triage and graph: TriageEngine (filter/dedupe/rank/FP-reduction), service-risk heuristic fallback, Neo4j graph persistence. **Comp: C4 — Completed**
W19	2026-07-20 to 2026-07-26	End-to-end integration: C1 to C2 to C3 to C4 chain wiring, shared JWT / Pydantic schemas, VulnerabilityClient and IRE execution orchestration. **Comp: ALL — Completed**
W20	2026-07-27 to 2026-08-02	IRE empirical evaluation design: 900-record campaign, metrics harness (intent accuracy, JSON validity, false-positive scope violations). **Comp: C1 — Completed**
W21	2026-08-03 to 2026-08-09	Evaluation execution: run campaign, collect results, analyse failure modes, fix discovered defects. **Comp: ALL — Completed**
W22	2026-08-10 to 2026-08-16	Hardening: adversarial prompt-injection defenses, hallucination detection, RBAC review, rate-limit verification. **Comp: C1 — Completed**
W23	2026-08-17 to 2026-08-23	AVAE verification: parser and ML scoring unit suites (52 tests), false-positive reduction metrics, model metric refinement. **Comp: C4 — Completed**
W24	2026-08-24 to 2026-08-30	Progress Presentation 2 preparation: cross-service integration test suite, prod docker-compose resource limits, QA and performance profiling. **Comp: ALL — Completed**

TABLE VI
TABLE 6.

Week	Date Range	Milestones and Deliverables
W25	2026-08-31 to 2026-09-06	**Progress Presentation 2 (31 Aug - 02 Sep).** Present all 4 components; finalize per-component SRS and combined system SRS (markdown, LaTeX, PDF) through **2026-09-05**. **Comp: ALL — Completed**
W26	2026-09-07 to 2026-09-13	Address PP2 feedback; finalize project timeline in MS Teams Planner; thesis outline and chapter map. **Comp: ALL — Upcoming**

TABLE VII
TABLE 7.

Week	Date Range	Milestones and Deliverables
W27	2026-09-14 to 2026-09-20	Thesis chapters 1-2: introduction, literature review; component documentation and data collection write-up. **Comp: ALL — Upcoming**
W28	2026-09-21 to 2026-09-27	Thesis chapters 3-4: system design and implementation per component; architecture figures and tables. **Comp: ALL — Upcoming**
W29	2026-09-28 to 2026-10-04	Thesis chapter 5: evaluation results and discussion; website content assembly and screenshots. **Comp: ALL — Upcoming**
W30	2026-10-05 to 2026-10-11	**Draft Thesis submission (11 Oct); Website Submission (11 Oct).** Finalize draft thesis and submit website. **Comp: ALL — Due 2026-10-11**

TABLE VIII
TABLE 8.

Week	Date Range	Milestones and Deliverables
W31	2026-10-12 to 2026-10-18	**Final Check List submission (14 Oct).** Complete final check list, logbook finalization, viva preparation. **Comp: ALL — Due 2026-10-14**
W32	2026-10-19 to 2026-10-25	**Final Presentation and Viva (19-21 Oct); Website evaluation and Logbook submission (19-21 Oct); Research Paper submission (23 Oct).** Deliver viva, submit logbook and research paper. **Comp: ALL — Due 2026-10-23**
W33	2026-10-26 to 2026-11-01	**Final Thesis Submission (28 Oct).** Incorporate viva feedback, finalize and submit thesis. **Comp: ALL — Due 2026-10-28**

TABLE IX
TABLE 9.

Week	Date Range	Milestones and Deliverables
W34	2026-11-02 to 2026-11-08	Paper revision: incorporate reviewer / supervisor feedback, prepare camera-ready version. **Comp: ALL — Upcoming**
W35	2026-11-09 to 2026-11-15	Publication follow-up: submit camera-ready, track acceptance and DOI assignment. **Comp: ALL — Upcoming**
W36	2026-11-16 to 2026-11-22	Publication evidence compilation: acceptance letter, DOI / published version, repository links. **Comp: ALL — Upcoming**
W37	2026-11-23 to 2026-11-29	Evidence package review: QA publication evidence bundle, align with supervisor. **Comp: ALL — Upcoming**
W38	2026-11-30 to 2026-12-06	**Research Paper publication evidence submission (01 Dec).** Submit evidence to faculty. **Comp: ALL — Due 2026-12-01**